

Hanbyul Joo

Associate Professor
Department of Computer Science and Engineering
Seoul National University

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EDUCATION

- Carnegie Mellon University** 2018
Ph.D. in Robotics Institute, School of Computer Science
Advisor: Prof. Yaser Sheikh
Thesis: [Sensing, Measuring, and Modeling Social Signals in Nonverbal Communication](#)
Thesis Committee: Yaser Sheikh (CMU), Takeo Kanade (CMU), Louis-Philippe Morency (CMU),
David Forsyth (UIUC), Mina Cikara (Harvard)
- KAIST** 2009
M.S. in Electrical Engineering
Advisor: Prof. In So Kweon
Thesis: Graph-based Boundary Matching for Deformable Objects
- KAIST** 2007
B.S. in Computer Science

WORK EXPERIENCES

- Seoul National University, Korea**
Associate Professor 2025 – Current
Assistant Professor Mar. 2022 – 2025
- Facebook AI Research (Meta AI), USA**
Research Scientist Jan. 2019 – Jan. 2022
- Oculus Research Pittsburgh, USA**
Research Intern May. 2017 – Oct. 2017
Worked on a 3D Human Body, Face, and Hand Tracking and Modeling project
- Disney Research Zurich, Switzerland**
Research Intern June. 2015 – Oct. 2015
Mentor: Thabo Beeler and Derek Bradley
Worked on a 3D Face Capture project
- Electronics and Telecommunications Research Institute (ETRI), South Korea**
Researcher Feb. 2009 – Jul. 2012
Worked on full 3D reconstruction technology for broadcasting communication fusion project
Developed a real-time markerless motion capture system using 20 broadcast cameras
Developed a system for automatic rigging and animation of 3D virtual avatar

PUBLICATIONS

- [1] [“Dexterous World Models”](#)
Byungjun Kim, Taeksoo Kim, Junyoung Lee, **Hanbyul Joo**
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
- [2] [“Vanast: Virtual Try-On with Human Image Animation via Synthetic Triplet Supervision”](#)
Hyunsoo Cha, Wonjung Woo, Byungjun Kim, **Hanbyul Joo**
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2026 ([Highlight](#)).
- [3] [“Text-Guided 6D Object Pose Rearrangement via Closed-Loop VLM Agents”](#)
Sangwon Baik, Gunhee Kim, Mingi Choi, **Hanbyul Joo**

European Conference on Computer Vision (ECCV), 2026.

- [4] "Target-Aware Video Diffusion Models"
Taeksoo Kim, **Hanbyul Joo**
International Conference on Learning Representations (ICLR), 2026.
- [5] "Durian: Dual Reference Image-Guided Portrait Animation with Attribute Transfer"
Hyunsoo Cha, Byungjun Kim, **Hanbyul Joo**
International Conference on Learning Representations (ICLR), 2026.
- [6] "GraspDiffusion: Synthesizing Realistic Whole-body Hand-Object Interaction"
Patrick Kwon, Chen Chen, **Hanbyul Joo**
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026.
- [7] "Audio Driven Real-Time Facial Animation for Social Telepresence"
Jiye Lee, Chenghui Li, Linh Tran, Shih-En Wei, Jason Saragih, Alexander Richard, **Hanbyul Joo**[†], Shaojie Bai[†] ([†] co-corresponding authors)
SIGGRAPH Asia, 2025.
- [8] "Learning to Generate Human-Human-Object Interactions from Textual Descriptions"
Jeonghyeon Na, Sangwon Baik, Inhee Lee, Junyoung Lee, **Hanbyul Joo**
Annual Conference on Neural Information Processing Systems (NeurIPS), 2025.
- [9] "DAViD: Modeling Dynamic Affordance of 3D Objects using Pre-trained Video Diffusion Models"
Hyeonwoo Kim, Sangwon Baik, **Hanbyul Joo**
International Conference on Computer Vision (ICCV), 2025.
- [10] "Learning 3D Object Spatial Relationships from Pre-trained 2D Diffusion Models"
Sangwon Baik, Hyeonwoo Kim, **Hanbyul Joo**
International Conference on Computer Vision (ICCV), 2025.
- [11] "HairCUP: Hair Compositional Universal Prior for 3D Gaussian Avatars"
Byungjun Kim, Shunsuke Saito, Giljoo Nam, Tomas Simon, Jason Saragih, **Hanbyul Joo**[†], Junxuan Li[†] ([†] co-corresponding authors)
International Conference on Computer Vision (ICCV), 2025. (Oral).
- [12] "ParaHome: Parameterizing Everyday Home Activities Towards 3D Generative Modeling of Human-Object Interactions"
Jeonghwan Kim*, Jisoo Kim*, Jeonghyeon Na, **Hanbyul Joo** (* co-first authors)
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
- [13] "PERSE: Personalized 3D Generative Avatars from A Single Portrait"
Hyunsoo Cha, Inhee Lee, **Hanbyul Joo**
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
- [14] "Beyond the Contact: Discovering Comprehensive Affordance for 3D Objects from Pre-trained 2D Diffusion Models"
Hyeonwoo Kim, Sookwan Han, Patrick Kwon, **Hanbyul Joo**
European Conference on Computer Vision (ECCV), 2024. (Oral)
- [15] "GALA: Generating Animatable Layered Assets from a Single Scan"
Taeksoo Kim, Byungjun Kim, Shunsuke Saito, **Hanbyul Joo**
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
- [16] "Mocap Everyone Everywhere: Lightweight Motion Capture With Smartwatches and a Head-Mounted Camera"
Jiye Lee, **Hanbyul Joo**
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
- [17] "Guess The Unseen: Dynamic 3D Scene Reconstruction from Partial 2D Glimpses"
Inhee Lee, Byungjun Kim, **Hanbyul Joo**
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
- [18] "PEGASUS: Personalized Generative 3D Avatars with Composible Attributes"
Hyunsoo Cha, Byungjun Kim, **Hanbyul Joo**
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024.

- [19] "CHORUS: Learning Canonicalized 3D Human-Object Spatial Relations from Unbounded Synthesized Images"
Sookwan Han, **Hanbyul Joo**
International Conference on Computer Vision (ICCV), 2023 (Oral).
- [20] "Chupa: Carving 3D Clothed Humans from Skinned Shape Priors using 2D Diffusion Probabilistic Models"
Byungjun Kim*, Patrick Kwon*, Kwangho Lee, Myunggi Lee, Sookwan Han, Daesik Kim, **Hanbyul Joo**
International Conference on Computer Vision (ICCV), 2023 (Oral).
- [21] "NCHO: Unsupervised Learning for Neural 3D Composition of Humans and Objects"
Taeksoo Kim, Shunsuke Saito, **Hanbyul Joo**
International Conference on Computer Vision (ICCV), 2023.
- [22] "Locomotion-Action-Manipulation: Synthesizing Human-Scene Interactions in Complex 3D Environments"
Jiye Lee, **Hanbyul Joo**
International Conference on Computer Vision (ICCV), 2023.
- [23] "BANMo: Building Animatable 3D Neural Models from Many Casual Videos"
Gengshan Yang, Minh Vo, Natalia Neverova, Deva Ramanan, Andrea Vedaldi, **Hanbyul Joo**
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022 (Oral)
- [24] "Ego4D: Around the World in 3,000 Hours of Egocentric Video"
Kristen Grauman et al.
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022 (Oral)
[Best Paper Award Finalist]
- [25] "Learning to Listen: Modeling Non-Deterministic Dyadic Facial Motion"
Evonne Ng, **Hanbyul Joo**, Liwen Hu, Hao Li, Trevor Darrell, Angjoo Kanazawa, Shiry Ginosar
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022.
- [26] "FrankMocap: A Fast Monocular 3D Hand and Body Motion Capture by Regression and Integration"
Yu Rong, Takaaki Shiratori, **Hanbyul Joo**
International Conference on Computer Vision (ICCV) Workshop, 2021.
- [27] "Exemplar Fine-Tuning for 3D Human Model Fitting Towards In-the-Wild 3D Human Pose Estimation"
Hanbyul Joo, Natalia Neverova, Andrea Vedaldi
International Conference on 3D Vision 2021 (3DV), 2021 (Oral).
- [28] "Body2Hands: Learning to Infer 3D Hands from Conversational Gesture Body Dynamics"
Evonne Ng, Shiry Ginosar, Trevor Darrell, **Hanbyul Joo**
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2021.
- [29] "3D Multi-bodies: Fitting Sets of Plausible 3D Human Models to Ambiguous Image Data"
Benjamin Biggs, Seb Ehrhardt, **Hanbyul Joo**, Ben Graham, Andrea Vedaldi, David Novotny
Conference on Neural Information Processing Systems (NeurIPS), 2020 (Spotlight).
- [30] "Perceiving 3D Human-Object Spatial Arrangements from a Single Image in the Wild"
Jason Y. Zhang*, Sam Pepose*, **Hanbyul Joo**, Deva Ramanan, Jitendra Malik, Angjoo Kanazawa
European Conference on Computer Vision (ECCV), 2020.
- [31] "PIFuHD: Multi-Level Pixel-Aligned Implicit Function for High-Resolution 3D Human Digitization"
Shunsuke Saito, Tomas Simon, Jason Saragih, **Hanbyul Joo**
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2020 (Oral).
- [32] "You2Me: Inferring Body Pose in Egocentric Video via First and Second Person Interactions"
Evonne Ng, Donglai Xiang, **Hanbyul Joo**, Kristen Grauman
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2020 (Oral).
- [33] "Single-Network Whole-Body Pose Estimation"
Gines Hidalgo, Yaadhav Raaj, Haroon Idrees, Donglai Xiang, **Hanbyul Joo**, Tomas Simon, Yaser Sheikh
International Conference on Computer Vision (ICCV), 2019.
- [34] "Towards Social Artificial Intelligence: Nonverbal Social Signal Prediction in A Triadic Interaction"
Hanbyul Joo, Tomas Simon, Mina Cikara, Yaser Sheikh
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2019 (Oral).
- [35] "Monocular Total Capture: Posing Face, Body and Hands in the Wild"

- Donglai Xiang, **Hanbyul Joo**, Yaser Sheikh
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2019 (Oral).
- [36] "Total Capture: A 3D Deformation Model for Tracking Faces, Hands, and Bodies"
Hanbyul Joo, Tomas Simon, and Yaser Sheikh
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2018 (Oral).
[Best Student Paper Award]
- [37] "Panoptic Studio: A Massively Multiview System for Social Interaction Capture"
Hanbyul Joo, Tomas Simon, Xulong Li, Hao Liu, Lei Tan, Lin Gui, Sean Banerjee, Timothy Godisart, Bart Nabbe, Iain Matthews, Takeo Kanade, Shohei Nobuhara, and Yaser Sheikh
Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2017.
- [38] "Hand Keypoint Detection in Single Images using Multiview Bootstrapping"
Tomas Simon, **Hanbyul Joo**, Iain Matthews, and Yaser Sheikh
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017.
- [39] "Panoptic Studio: A Massively Multiview System for Social Motion Capture"
Hanbyul Joo, Hao Liu, Lei Tan, Lin Gui, Bart Nabbe, Iain Matthews, Takeo Kanade, Shohei Nobuhara and Yaser Sheikh
International Conference on Computer Vision (ICCV), 2015 (Oral).
- [40] "MAP Visibility Estimation for Large-Scale Dynamic 3D Reconstruction"
Hanbyul Joo, Hyun Soo Park, and Yaser Sheikh
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2014 (Oral).
- [41] "Graph-based Shape Matching for Deformable Objects"
Hanbyul Joo, Yekeun Jeong, Olivier Duchenne, and InSo Kweon
IEEE International Conference on Image Processing (ICIP), 2011.
- [42] "Graph-Based Robust Shape Matching for Robotic Application"
Hanbyul Joo, Yekeun Jeong, Olivier Duchenne, Seong-Young Ko, and InSo Kweon
IEEE International Conference on Robotics and Automation (ICRA), 2009.

AWARDS & SCHOLARSHIPS

CVPR Outstanding Area Chair Award	2026
Okawa Research Grant Award	2025
Excellence in Teaching Award from SNU College of Engineering	2023, 2024
CVPR Best Student Paper Award Out of 979 accepted papers (out of 3309 submissions)	2018
Samsung Scholarship Tuition and stipend for Ph.D. study (\$50K/year, for 5 years)	2012 - 2017
Governmental Scholarship for KAIST Graduate Students	2007 - 2009
Governmental Scholarship for KAIST Undergraduate Students	2002 - 2006

TUTORIAL/WORKSHOP ORGANIZATION

- "3rd Workshop on Dexterous Manipulation: Learning and Control with Diverse Data"
In Conjunction with RSS 2024.
- "Artificial Social Intelligence"
In Conjunction with ECCV 2024, ICCV 2023, and CVPR 2022.
- "Virtual Try-On"
In Conjunction with CVPR 2024.
- "Visual Recognition for Images, Video, and 3D"
In Conjunction with ECCV 2020.

"DIY A Multiview Camera System: Panoptic Studio Teardown "
In Conjunction with **CVPR 2017**.

ACADEMIC SERVICES

Lead Area Chair , CVPR, ECCV	2026
Area Chair , CVPR	2022-2025
Area Chair , ICCV	2023, 2025
Area Chair , NeurIPS	2023-2026
Area Chair , ICLR	2026
Area Chair , 3DV	2025, 2026
Area Chair , CoRL	2026
Area Chair , WACV	2023
Technical Paper Committee , SIGGRAPH Asia	2022

DATASETS & OPENSOURCE

- ParaHome**: <https://github.com/snuvclab/ParaHome>
3D Dataset for Human-Object Interaction Captures
- FrankMocap**: <https://github.com/facebookresearch/frankmocap>
A Strong and Easy-to-use Single View 3D Hand+Body Pose Estimator
- PIFuHD**: <https://github.com/facebookresearch/pifuhd>
High-Resolution 3D Human Digitization Code from A Single Image.
Top 1 in Trending repository list on GitHub
9.4K stars and 1.4K forks (as of Mar 2024)
- Panoptic Studio Dataset**: <http://domedb.perception.cs.cmu.edu>
A dataset of 3D hands, bodies, and face motion for social groups captured by the Panoptic Studio
- OpenPose**: <https://github.com/CMU-Perceptual-Computing-Lab/openpose>
The first real-time multi-person system to jointly detect body, hand, and facial keypoints on single images
Top 1 in Trending repository list on GitHub
29.6K stars and 7.8K forks (as of Mar 2024)

SELECTED PRESS COVERAGE

- Arirang TV**, *AI: Reading Humans*, Aug. 2025
- Arirang TV**, *The Evolution of AI*, Jul. 2025
- FOSSBYTES**, Facebook's New "PiFuHD" Tech Can Construct 3D Models From Photos, Jun. 2020
- IT Media (Japanese)**, High-definition 3D model creation from one portrait with AI Developed such as Facebook, Jun. 2020
- WIRED**, Inside the Panoptic Studio, the Dome That Could Give Robots Super-Senses, Feb. 2018
- BBC News**, The Dome Which Could Help Machines Understand Behavior, Oct. 2017
- Reuters**, 500-Camera Dome Trains Computer To Read Body Language, Oct. 2017
- EBS (Korean TV Channel)**, Docuprime: The Global War For Talent, Mar. 2017
- CMU News**, Scientists Put Human Interaction Under The Microscope, Mar. 2017
- The Verge**, Cracking The Elaborate Code, Dec. 2016

SPIEGEL ONLINE, [The Panoptic Studio: Computer Decipher The Secrets of Body Language](#), Dec. 2015

Wired (Italian), [Panoptic Studio: The Latest Generation of Motion Capture](#), Jul. 2015

Voice of America, [New Studio Yields Most Detailed Motion Capture in 3D](#), Apr. 2015

Reuters, [Motion capture on a whole new level](#), Apr. 2015

Discovery Channel Canada, [Daily Planet Show, Future Tech: Panoptic Studio](#), Jan. 2015

IEEE Spectrum, [Camera-Filled Dome Recreates Full 3-D Motion Scenes](#), Jul. 2014

Discovery News, [Amazing 3-D Flicks from Dome of 500 Cameras?](#), Jul. 2014

NBC NEWS, [Camera-Studded Dome Tracks Your Every Move With Precision](#), Jul. 2014

CNet, [Tomorrow Daily: New video capture tech](#), Jul. 2014

Engadget, [Watch A Dome Full of Cameras Capture 3D Motion in Extreme Detail](#), Jul. 2014

GIZMODO, [A Dome Packed With 480 Cameras Captures Detailed 3D Images In Motion](#), Jul. 2014

THE Verge, [Scientists build a real Panopticon that captures your every move in 3D](#), Jul. 2014

Science Daily, [Hundreds of Videos Used To Reconstruct 3-D motion Without Markers](#), Jul. 2014

PHYS.ORG, [Researchers Combine Hundreds of Videos To Reconstruct 3D Motion Without Markers](#), Jul. 2014

Slate, [Freezing Memories in Time](#), Jul. 2014

PetaPixel, [Researchers Use a 480-Camera Dome to More Accurately Capture 3D Motion](#), Jul. 2014

Gizmag, [Camera-studded Dome Used To Reconstruct 3D Motion](#), Jul. 2014

theENGINEER, [3D Motion Captured Without Markers](#), Jul. 2014

CMU News, [Carnegie Mellon Combines Hundreds of Videos To Reconstruct 3D Motion ...](#), Jul. 2014

PATENTS

Motion capture apparatus and method (Patent No.: US 8805024 B2)

Hanbyul Joo, Seong-Jae Lim, Ji-Hyung Lee, Bon-Ki Koo

Method for automatic rigging and shape surface transfer of 3D standard mesh model based on muscle and nurbs by using parametric control (Patent No.: US 7171060 B2)

Seong Jae Lim, Ho Won Kim, **Hanbyul Joo**, Bon Ki Koo

3D model shape transformation method and apparatus (Patent No.: US8922547B2)

Seong-Jae Lim, **Hanbyul Joo**, Seung-Uk Yoon, Ji-Hyung Lee, Bon-Ki Koo.

Student/Intern Mentorship

[Subin Jeon](#), Postdoc in SNU (2026 - present)

Byungjun Kim, MS/PhD in SNU (Fall 2022 - present)

Jeonghwan Kim, MS/PhD in SNU (Fall 2022 - present)

Hyunsoo Cha, MS/PhD in SNU (Fall 2022 - present)

Jisoo Kim, MS/PhD in SNU (Fall 2022 - present)

Taeksoo Kim, MS/PhD in SNU (Spring 2023 - present)

Inhee Lee, MS/PhD in SNU (Spring 2023 - present)

Sangwon Baik, MS/PhD in SNU (Fall 2023 - present)

Hyeonwoo Kim, MS/PhD in SNU (Spring 2024 - present)

Mingi Choi, MS/PhD in SNU (Spring 2025 - present)

Junyoung Lee, MS in SNU (Spring 2025 - present)

Kyungwon Cho, MS/PhD in SNU (Spring 2026 - present)

Gunhee Kim, MS/PhD in SNU (Spring 2026 - present)

Wonjung Woo, MS/PhD in SNU (Spring 2026 - present)

Jongbin Lim, MS/PhD in SNU (Spring 2026 - present)

Taeyun Ha, MS/PhD in SNU (Fall 2026 - present)

Chae-Yeon Lee, PhD in SNU (Fall 2026 - present)

Jeonghyeon Na, MS/PhD in SNU (Fall 2023 - Spring 2026)

Jiye Lee, MS/PhD in SNU (Fall 2022 - Dec 2024)

Yonwoo Choi, MS in SNU (Spring 2022 - Feb 2024)

Evonne Ng, PhD in UC Berkeley
FAIR-BAIR program mentor (Spring 2020 - Fall 2021)

Xiang Xu, PhD in Simon Fraser University
FAIR internship mentor (Summer 2020 - Fall 2020)

Yu Rong, PhD in The Chinese University of Hong Kong
FAIR internship mentor (Spring 2020)

Shunsuke Saito, PhD in University of Southern California
FAIR internship mentor (Summer 2019)

Donglai Xiang, MS in Carnegie Mellon University
Mentoring for [Monocular Total Capture Project](#)

TEACHING

Instructor , Seoul National University Artificial Intelligence (4190.408)	<i>Fall 2022-2025</i>
Instructor , Seoul National University 3D Computer Vision (M1522.007100)	<i>Spring 2022-2025</i>
Instructor , Seoul National University Computer Vision (M1522.001000)	<i>Spring 2025</i>
Co-Instructor , Seoul National University Theories and Lab of VR/AR (M2177.007800)	<i>Spring/Fall 2022-2025</i>
Guest Lecturer , Carnegie Mellon University Computational Photography (15-463) (Instructor: Ioannis (Yannis) Gkioulekas)	<i>Fall 2018</i>
Guest Lecturer , Carnegie Mellon University Computer Vision (16-720) (Instructor: Kris Kitani)	<i>Spring 2018</i>
Teaching Assistant , Carnegie Mellon University Computer Vision (16-720) (Instructor: Martial Hebert)	<i>Fall 2014</i>